WORKED SOLUTION

STEP I 2000, Question 1

Pure • Logarithms and Leading Digits

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q1

The key idea

The leading digit of a large number N is decided entirely by the *fractional part* of $\log_{10} N$. If $\log_{10} N = k + f$ with k an integer and $0 \leq f < 1$, then $N = 10^f \times 10^k$, and the first digit of N is d exactly when $\log_{10} d \leq f < \log_{10}(d+1)$. So all we need are the logs of the digits 1 to 9, which come from $\log_{10} 2$ and $\log_{10} 3$, and a careful eye on how much rounding error we can afford.

Solution

Throughout, log means $\log_{10}$. The interval $[0,1)$ splits into nine pieces, one for each possible leading digit:



A fractional part f landing in the piece labelled d gives leading digit d.

Part (i)

Since $5 = 10/2$ and $6 = 2 \times 3$,

$$\log 5 = 1 - \log 2 = 1 - 0.301029996 = 0.698970004 \approx 0.699,$$
$$\log 6 = \log 2 + \log 3 = 0.301029996 + 0.477121255 = 0.778151251 \approx 0.778,$$

each to three decimal places.

Now

$$\log 3^{100} = 100 \log 3 = 47.7121255.$$

The given value of $\log 3$ is correct to within 5×10^{-10}, so this is correct to within

5×10^{-8}; certainly $47.712 < \log 3^{100} < 47.713$. Comparing with

$$\log(5 \times 10^{47}) = 47 + \log 5 \approx 47.699 \quad \text{and} \quad \log(6 \times 10^{47}) = 47 + \log 6 \approx 47.778,$$

we have

$$\log(5 \times 10^{47}) < \log 3^{100} < \log(6 \times 10^{47}).$$

The margins (about 0.013 and 0.066) dwarf any rounding error, and log is strictly increasing, so

$$5 \times 10^{47} < 3^{100} < 6 \times 10^{47},$$

as required. Hence 3^{100} is $5.\ldots \times 10^{47}$: its first digit is $\boxed{5}$ (and it has 48 digits).

Part (ii)

We use the same idea with $\log 2 = 0.301029996$, which is in error by at most 5×10^{-10}. Multiplying by 10^n multiplies that error by 10^n, so we must check that the fractional part stays safely inside one piece of the number line.

2^{1000}. $\log 2^{1000} = 301.029996$, with error at most 5×10^{-7}. The fractional part, about 0.030, lies between $\log 1 = 0$ and $\log 2 \approx 0.301$, so $2^{1000} = 1.\ldots \times 10^{301}$ and its first digit is $\boxed{1}$.

$2^{10\,000}$. $\log 2^{10\,000} = 3010.29996$, with error at most 5×10^{-6}. The fractional part is between 0.299955 and 0.299965.

Ask yourself The fractional part 0.29996 is very close to $\log 2 = 0.30103$. Are we sure the first digit isn't 2?

Yes: the gap is $0.30103 - 0.29996 \approx 0.0011$, which is more than two hundred times the largest possible error of 5×10^{-6}. So the fractional part really is less than $\log 2$, and $10^{3010} < 2^{10\,000} < 2 \times 10^{3010}$. (Equivalently, $2^{10\,000} < 2 \times 10^{3010}$ says $2^{9999} < 10^{3010}$, i.e. $\log 2 < 3010/9999 = 0.3010301\ldots$, which is true.) The first digit is $\boxed{1}$.

$2^{100\,000}$. $\log 2^{100\,000} = 30102.9996$, with error at most 5×10^{-5}, so the fractional part lies between 0.99955 and 0.99965. We need to know which piece of the number line this is in, and the top piece starts at

$$\log 9 = 2 \log 3 = 0.954242510.$$

Since $0.954 < 0.99955$ and $0.99965 < 1$, we get $9 \times 10^{30102} < 2^{100\,000} < 10^{30103}$, and the first digit is $\boxed{9}$.

Ask yourself Couldn't the error push the fractional part past 1, making the number $1.\ldots \times 10^{30103}$?

Only if the true value of $100\,000 \log 2$ were at least 30103, which would need an error of 0.0004 in it, eight times more than the maximum 5×10^{-5}. So the first digit is 9 and $2^{100\,000}$ has 30 103 digits.

Checks

Exact integer arithmetic by computer confirms everything: 3^{100} has 48 digits and begins $515\ldots$; 2^{1000} begins $107\ldots$ (302 digits); $2^{10\,000}$ begins $199\ldots$ (3011 digits); and $2^{100\,000}$ begins $999\ldots$ (30 103 digits). A quick sanity check on the method: $\log 2^{10} = 3.0103$ has fractional part $0.0103 < \log 2$, matching $2^{10} = 1024$.

Common slips

- Reading the first digit from the integer part of the logarithm. The integer part only gives the number of digits (one more than it); the leading digit lives in the fractional part.
- Forgetting that multiplying by 100 000 also multiplies the rounding error in $\log 2$. Nine decimal places are just enough here, and it's worth saying why.
- Seeing $f = 0.9996 \approx 1$ and concluding the digit is 1. A fractional part just below 1 means the number is just *below* a power of ten, so it starts with 9.

Beyond STEP

The leading digits of 2^n are governed by the fractional parts of $n \log 2$. Because $\log 2$ is irrational, these fractional parts are *equidistributed* in $[0, 1)$ (Weyl's equidistribution theorem), so the proportion of powers of 2 starting with digit d tends to $\log(1 + 1/d)$: this is Benford's law, and about 30% of powers of 2 begin with a 1. For the irrationality and approximation side of the story, Davenport's *The Higher Arithmetic* is a lovely read.

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